Evaluation Report

Problem 10: Speech Transcript Vectorization using Bag-of-Words and Collocation Features

Student: ET23BTCO127 | All numbers come from `python -m src.pipeline` (seed 42) and can be reproduced.

1. Dataset

350 synthetic voice-assistant transcripts, 5 intents (Navigation, Media, Climate, Phone, Reminder), 70 per intent, generated by `scripts/make_dataset.py`. Each transcript was built from command templates, then corrupted to look like raw ASR output: no punctuation, random casing, spoken fillers (um, uh, you know, i mean), informal contractions (gonna, wanna, lemme), and occasional dropped or duplicated words (about 8% and 5%).

Important: because the data is template-generated, intents are easy to separate. The results below demonstrate that the pipeline works and show how the parameters behave; they should not be read as real-world accuracy.

2. Preprocessing statistics

Measure	Before	After	Reduction
Total tokens	2610	1127	56.8%
Vocabulary size	202	164	18.8%
Avg. tokens per transcript	7.46	3.22	-

Cleaning removed more than half of all tokens (fillers, stop words, politeness words) while the vocabulary shrank by about 19% (casing and inflection merged).

3. Feature matrix (default settings)

Defaults: min_df = 1, bigram collocations, min_freq = 5, PMI threshold = 3.0. Result: 350 x 227 matrix = 164 BoW columns + 63 collocation columns; 1558 non-zero entries; sparsity 98.04%; 4.45 non-zero values per transcript on average.

4. Top-20 extracted collocations (qualitative evaluation)

Rank	Collocation	Count	PMI
1	arrival_time	5	7.82
2	eight_fifteen	5	7.82
3	estimate_arrival	5	7.82
4	give_direction	5	7.82
5	last_number	5	7.82
6	redial_last	5	7.82
7	customer_service	6	7.55
8	skip_track	6	7.55
9	mute_radio	5	7.33
10	park_lot	5	7.33
11	defrost_windshield	8	7.14

Rank	Collocation	Count	PMI
12	run_late	8	7.14
13	shuffle_favorite	8	7.14
14	voice_mail	8	7.14
15	start_navigation	6	7.14
16	make_cooler	5	7.14
17	show_traffic	5	7.14
18	calendar_today	5	6.97
19	shop_list	5	6.97
20	avoid_toll	10	6.82

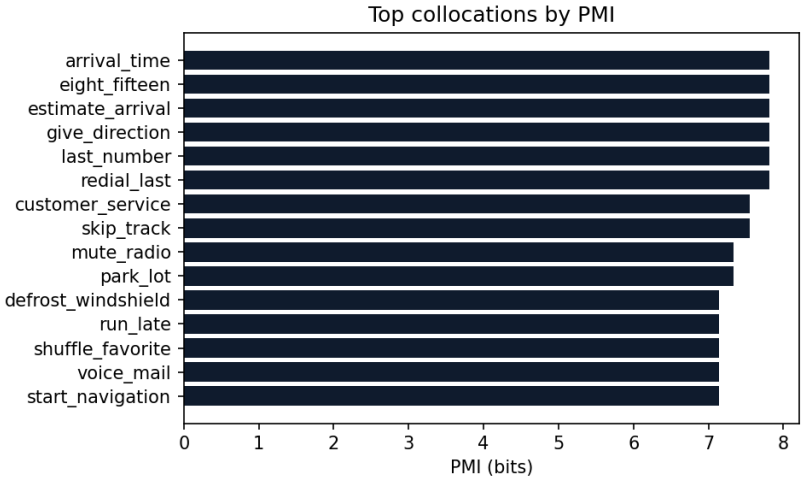


Figure 1. Highest-PMI collocations.

Assessment. Almost all of the top 20 are meaningful command phrases from the domain: *customer_service*, *skip_track*, *defrost_windshield*, *voice_mail*, *redial_last*, *start_navigation*, *avoid_toll*. Some pairs are fragments that exist only because stop words were removed between them (*give_direction* from "give me directions", *road_take* from "toll roads and take"). With min_freq = 2 the top of the list was dominated by one-off pairs such as *buy_milk* (count 2), which is exactly the rare-pair bias described in the methodology.

5. BoW versus augmented BoW

Classifier: linear SVM (scikit-learn LinearSVC). The vectorizer, including PMI scores, is fitted inside each fold on training data only. "5-fold" trains on 80% and tests on 20%. "Low-data" trains on only 20% of the transcripts and tests on the other 80%, a harder setting.

Configuration	BoW	Colloc.	Total	Sparsity %	5-fold acc.	5-fold macro-F1	Low-data acc.
BoW only	164	0	164	98.05	1.0000	1.0000	0.8864
BoW + ALL bigrams (no PMI filter)	164	236	400	98.65	1.0000	1.0000	0.8829
BoW + PMI bigrams	164	63	227	98.04	0.9971	0.9971	0.8864
BoW + PMI bigrams + trigrams	164	90	254	98.05	0.9971	0.9971	0.8864

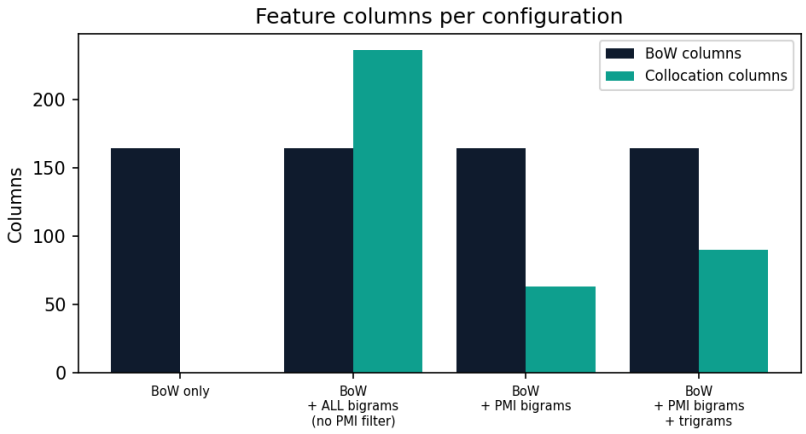


Figure 2. Feature columns per configuration.

Findings. (a) Collocation features did **not** improve intent accuracy on this dataset: BoW alone already reaches 100% in 5-fold CV, and in the low-data setting all four configurations are within about 1 percentage point (88.3% to 88.6% in the low-data runs), which is not a meaningful difference on 350 samples. This is expected, since single words such as "alarm", "call" or "volume" already identify the intent. (b) The PMI filter is, however, an effective **dimensionality control**: keeping all bigrams adds 236 columns, while PMI-filtered bigrams add 63 and give practically the same accuracy. (c) The augmented matrix is more **interpretable**: columns such as *credit_card* or *customer_service* are human-readable phrase features.

6. PMI threshold and min_freq sweeps

PMI threshold	Collocation cols	Total cols	5-fold acc.	Low-data acc.
-1	65	229	0.9971	0.8864
0	65	229	0.9971	0.8864
1	65	229	0.9971	0.8864
2	64	228	0.9971	0.8864
3	63	227	0.9971	0.8864
4	60	224	0.9971	0.8864
5	46	210	1.0000	0.8864
6	36	200	1.0000	0.8864
7	17	181	1.0000	0.8864

Raising the threshold from -1 to 7 shrinks the collocation block from 65 to 17 columns with no loss of accuracy (differences are within one transcript out of 350). The min_freq sweep (Architecture document, Section 8) shows a much stronger effect on width: 214, 147, 100, 63 and 20 collocation columns for min_freq 1, 2, 3, 5 and 8.

7. Failure analysis

- **Artificial adjacency.** "where is the closest gas station" becomes closest gas station after stop-word removal, so *closest* and *gas* become adjacent and can be counted as a pair although the speaker never said them together. Remedy: extract collocations before stop-word removal, or only join tokens that were adjacent in the original transcript.
- **Frequent but low-PMI phrases.** "turn up" (18 occurrences) has PMI 2.98 and falls below the default threshold, while rarer exclusive pairs rank higher. PMI measures exclusivity, not importance. Remedy: combine PMI with a frequency or chi-square / likelihood-ratio measure.

- **ASR deletions and duplications.** A dropped word (about 8% of transcripts) breaks a phrase, so its collocation column stays 0 although the BoW columns still fire. This is a real advantage of keeping both blocks.
- **Lemmatizer side effects.** WordNet may over-reduce words ("conditioning" to "condition", "media" to "medium"); a small exception list handles the known cases but others may exist.
- **Small-sample variance.** With 350 transcripts, a difference of 1% is 3 or 4 transcripts. No claim of improvement is made.

8. Conclusions

- The pipeline turns noisy transcripts into a compact, interpretable BoW + collocation matrix and exposes all dimensionality controls through the web app.
- A minimum-frequency filter (>= 3, default 5) is essential for PMI; the PMI threshold then fine-tunes the width of the collocation block.
- On the synthetic dataset, collocation features keep accuracy essentially unchanged (99.7% versus 100% in 5-fold CV, i.e. one transcript) while adding readable phrase features; they do not raise accuracy because single words are already decisive.
- Future work: real ASR transcripts, chi-square and likelihood-ratio measures (Problem 17), TF-IDF weighting, and collocations extracted before stop-word removal.